

Lalith Kumar A Gen AI Engineer

✉ adhimulamlalith22@gmail.com ☎ +91 9840403254

🌐 [linkedin.com/in/a-lalith-kumar-66606517a](https://www.linkedin.com/in/a-lalith-kumar-66606517a)

👤 SUMMARY

Innovative and results-driven LLM/GenAI Engineer with hands-on experience designing and deploying intelligent systems powered by Agentic AI architectures, RAG pipelines, and custom deep learning models. Proven success in building modular multi-agent workflows, retrieval systems, and real-time inference APIs across insurance and computer vision domains. Skilled in fine-tuning LLMs and architecting scalable RAG and agentic solutions. Experienced in integrating Azure OpenAI and Azure Document Intelligence to enhance knowledge retrieval, document understanding, and enterprise automation. Adept at transforming complex requirements into production-grade AI applications that reduce manual effort, improve decision-making, and accelerate business processes.

🧰 WORK EXPERIENCE

Gen AI Engineer, Sigmasoft Infotech

07/2025 – Present | Hyderabad

Agentify

- Designed and implemented a modular platform for configuring autonomous **agents** and **multi-agent** swarms, enabling reusable workflows across diverse business use cases.
- Added functionality to create dynamic pipelines for building graph db, sql db, and vector db for agents to access.
- Enabled low-friction creation of new agents and **swarms** through declarative configuration, reducing manual engineering effort and accelerating experimentation and go-live cycles.
- Create a Flow Engine along with nodes that enables user to create workflows that can be triggered and scheduled.

Sales Copilot

- Developed an agentic sales-intelligence system using **Autogen**, combining a Structured Query Language (SQL)-based agent, a **Retrieval-Augmented Generation (RAG)** agent, and a supervisor business-logic agent.
- Automated query understanding, SQL generation, table-level retrieval, and business-ready summarisation without manual intervention.
- Could effectively conduct analysis on retrieved data and infer on the data.
- Improved access to sales metrics and reduced ad-hoc analysis workload for business teams.

Project Engineer, Wipro Ltd.

06/2021 – 07/2025 | Hyderabad, India

Agentic Triage System

- Built and maintained an agentic triage system using **Hugging Face**, **LangChain**, **LangGraph**, to handle insurance-related queries from policy forms using **Azure Document Intelligence**.
- Designed modular agents that can manage tasks across sub agents, summarise and route while effectively calling tools.
- **Chain of Thought** is implemented to optimize the routing between agents while **ReAct** is used with agents that have tools.
- The images are preprocessed and are passed to **Azure AI vision** for damage verification and fraud signal detection.
- Used **LangSmith** to actively monitor and create **datasets** for fine tuning **LLMs**.
- Improved triage speed and consistency, reducing manual operational effort.

Knowledge Base & Zero-Shot RAG Chatbot

- Created a scalable **Python**-based knowledge-base system that accelerated development cycles by 35 percent.

- Integrated **Hugging Face transformers** and **Autogen** to build a **Retrieval-Augmented Generation (RAG)** chatbot using the **Mistral-7B** Instruct model.
- **Zero shot prompting** with proper intent and negations could effectively respond to the users.
- Reduced cost incurred per query using advanced prompting techniques, **guardrails** and **caching**.
- Improved document-retrieval accuracy and reduced resolution effort through automated contextual Q&A.

Computer Vision Claims Automation

- Developed a **VGG-19** based computer-vision model to classify damage severity, cause, and impact from customer images.
- Deployed the workflow on **Amazon Web Services (AWS) SageMaker** with a continuous integration/continuous delivery (CI/CD) setup using **MLflow** and **GitHub Actions**.
- Improved model accuracy by 30 percent and reduced claims-processing effort through automated evaluation.

SKILLS

Frameworks/ Libraries: Langchain, Langgraph, Autogen, Pydantic, Prefect, Fast API, PyTorch, Hugging Face, Accelerate, Pandas, Langsmith, DeepEval

Gen AI Engineering: Prompt Engineering, RAG, Agentic AI, Structured Outputs, Function/Tool Calling, Map reduce, Orchestrating pipelines, Reranking, Cross Encoders

LLM Evaluation: LLM as Judge, Human Evaluation, Metric Based

Cloud Stack: Azure Functions, Azure AI Foundry, Azure Document Intelligence, Amazon Bedrock, Amazon Sagemaker

Similarity Search: ChromaDB, FAISS, Postgres Pgvector

Miscellaneous: Sqlite3, CI/CD, ONNX inference, Git, Docker

Optimizations: LoRa, QLoRa, Prompt Tuning, K3S

PROJECTS

Terra Scope

01/2026

Engineered TerraScope geospatial AI platform analyzing single to multiple locations via Google Earth Engine, OSMnx POIs, and nightlight economic proxies. It generates interactive infrastructure maps, population heatmaps, and investment radar charts for real estate feasibility, and creates an optional automated report.

Swing Trader

Developed Gen-AI swing trading bot with agentic frameworks for fundamental/quant analysis, TFT forecasting, and 4-hour market monitoring; covered 700+ NSE companies with peer comparisons, buy/sell signals, and crash alerts, cut analysis time 70% and boosted user profits.

Health Score Classification

Devised a real-time health assessment algorithm that synthesized critical patient metrics to compute an intuitive health score, facilitating proactive medical decisions.

Introduced an AI-powered scoring engine that translated complex clinical data into actionable insights, minimizing cognitive load for physicians.

Contributed to faster diagnostic workflows by reducing evaluation time and enabling more confident, data-backed treatment planning.

Auto Animator

Constructed a deep convolutional generative adversarial network to generate animation tensors (x, y, z, quaternion, bones, time), enabling realistic motion synthesis from skeletal input and temporal data.

Optimized a multi-layer architecture to autonomously generate dynamic, life-like animations, significantly reducing the need for manual rigging or frame-by-frame intervention.

EDUCATION

Bachelors in Electronics and Computer Science,
VIT University, Chennai

2017 – 2021

CERTIFICATIONS

- Aerial Robotics 
- Architecting with Google Compute Engine 